

Homework: running the full dataset and all the steps

In the workshop you processed a handful of images and a pretrained model to see how each step behaves.

This guide is for running the full training set on *Listeria* on your own computer, in your own time.

Feature extraction and model training on the full set takes a while.

You will run the same pipeline you have already run, just at full scale.

Before you start

- Download the full training set from Zenodo:
https://zenodo.org/records/20377151/files/17_11_2023_3D_z_21.zip?download=1
- Unzip it and keep all images in a single folder. It is a large file (~50 GB) — make sure you have enough free disk space before you begin.

Prepare your laptop for a long run. A full run writes a large feature table and must not be interrupted. Plug in the power, disable sleep, and postpone system updates for the duration. Start the MicroICS (Look document: **Installation guide**).

Generate features

This is the step to leave running overnight, or across two days, depending on how many parallel workers your machine supports. The software measures around 2,700 descriptors for every image, and on the full set that is a substantial amount of computation.

Do: Run feature extraction in labelled mode on the full image folder. How to do this is described in the “**Hands on exercise**” - Generate features.

Outputs:

When processing is complete, datafile.tsv contains one row per image. Open it in Excel and check the first column: every image should be listed by name. Any missing images indicate a naming or format problem with those specific files.

Train a model

Once datafile.tsv is complete, run the training.

Do: Run training. How to do this is described in the “**Hands-on exercise**” – Train a model. Use all learners to compare algorithms, or a single learner if you already know which one you want. For this dataset, random forest is the recommended choice.

Running all learners takes a while — comparable to feature extraction — so it is worth doing once, to see which method performs best, and thereafter running only the best-performing model.

Outputs:

This step produces the trained model and the performance results together.

The outputs to examine first are those from the workshop: `classification_all`, `rankings_label`, `rankings_date`, and `ablation_ranking_all`. Full descriptions of every output file are given below.

File	What it tells you
<code>classification_all</code>	Performance of each classifier. Use this to choose your classifier, and to check it beats the majority-class baseline.
<code>classification_no_counts</code>	The same, excluding count-based (abundance) features. The difference shows how much performance depends on abundance related features.
<code>ablation_ranking_all</code>	How many features are needed for top performance. Provided only for random forest.
<code>rankings_label</code>	Which structural features best separate your biological groups — the ranking that carries the biological meaning. Provided only for random forest.
<code>rankings_date</code>	Which features track the acquisition date rather than biology. Comparing this with <code>rankings_label</code> is the batch-effect check. Provided only for random forest.
feature heatmap	Shows how strongly the top features correlate with one another. Each cell gives the correlation between two features, so bright blocks along the diagonal reveal groups of features that are essentially measuring the same thing. Use it to see how many independent signals your top features really represent.
confusion matrix	Shows how often predictions are correct and, when they are wrong, which classes are mistaken for which. Rows represent the true class, columns the predicted class; the diagonal shows correct predictions, off-diagonal cells show errors. Classes that are repeatedly confused are structurally similar within the model, which indicates which mistakes are expected and helps you interpret low-confidence predictions in the right context.

Inference

Inference is two steps: first calculate features from the new images, then run the model on them.

Inference images (Zenodo):

https://zenodo.org/records/20390961/files/E8_images_for_prediction.zip?download=1

Do:

Run (1) “generate features” on the downloaded inference images in unlabelled mode. How to do this is described in the “**Hands-on exercise**” – Inference part 1.

Run (2) inference on the resulting features. How to do this is described in the “**Hands-on exercise**” – Inference part 2.

Outputs:

Check `inference_summary` first — unexpected or strange class names mean the labels were not read as intended.

Full output descriptions are given below:

File	What it tells you
inference_summary	Overview of the run — model used, number of classes, predictions made. Check first; strange class names mean the labels were not read as intended.
rf_predictions	Predicted class per image. The primary result.
rf_probabilities	Confidence behind each prediction. Near 1 = clear match; low = the image resembles more than one class, or a structure not in the training set. A low score is informative, not a failure.
rf_features	The feature values calculated for the new images. If external features were added, check here that they were read correctly.
rf_feature_importance	Feature importance for the inference model.
rf_shap_values	Per-image feature attributions.
rf_summary_plot	Visual (SHAP) summary of which features drove the predictions.